Japan: BoJ not solely responsible for worsening JPY real purchasing power

DESKNOTE
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OUR KEY VIEWS

- The long-term decline of the yen's real purchasing power is a structural issue partly caused by the obstruction of the Balassa-Samuelson effect (the decoupling of productivity and wages), rather than merely being a result of Bank of Japan monetary policy.
- While the BoJ's gradual monetary policy normalisation has been negative for the yen, the currency has now been weakening for more than a quarter of a century, reflecting the structural downward pressure associated with declines in Japan's relative wage and price levels, as domestic firms have repeatedly refused to reward workers for their improving productivity.
- Absent a scenario where Japan finds itself alone in experiencing high wage growth and inflation, we do not expect more than a modest appreciation of the yen in real terms, even if Japan's interest rate differentials (versus the US and Europe) narrow due to continued BoJ monetary policy normalisation.

Despite recent defensive interventions by Japanese authorities, the JPY remains near 40-year lows. Meanwhile, the real effective exchange rate – a weighted average relative to a basket of foreign trading partners' currencies, adjusted for price level differences – has fallen below levels seen at the start of the 1970s (Figure 1). Japan-versus-overseas real interest rate differentials are not enough to explain this REER weakness, in our view (Figure 2). We believe another factor is likely behind the sharp cheapening of Japanese goods and services.

What is the Balassa-Samuelson effect? Economists often cite this effect when discussing the long-term decline in the yen's REER. Essentially, a currency's real purchasing power is likely to strengthen if improved productivity in the goods sector raises wages there. Through labour market mechanisms, these wage hikes then spread to the services sector. Because services are not as productive as goods, these increased wages drive up services prices more than goods prices, leading to higher overall domestic inflation.

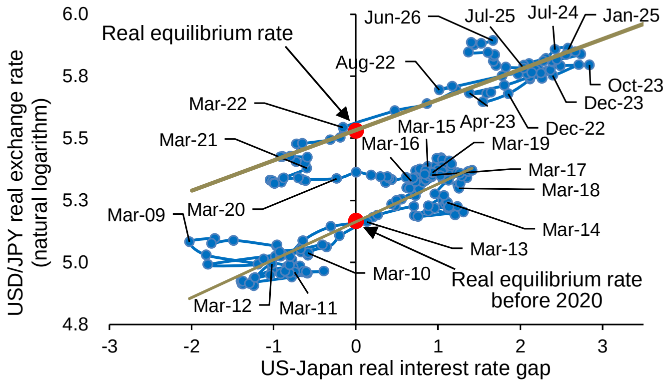
Many experts have cited this effect as a primary cause of the yen's continued real appreciation through the mid-1990s. However, what explains the depreciation since then? Some contend that the JPY REER dropped due to a lack of productivity growth in the goods sector and a corresponding lack of wage growth in both the goods and services sectors, but we are not convinced by this 'reverse Balassa-Samuelson effect' argument.

Japanese productivity has, in fact, kept improving: Hourly productivity has risen by about 30% over the past 25 years in Japan, more than in Germany and France, but admittedly less than in the US (Figure 3). During the same period, however, the hourly real wage has remained stagnant (Figure 4).

Fig. 1: Real effective exchange rate – JPY (CY2010 = 100)

Sources: BoJ, BNP Paribas

Fig. 2: US–Japan real interest rate gap versus USDJPY real exchange rate



Sources: MIC, BLS, Bloomberg, BNP Paribas

Our hypothesis is that this decoupling of productivity and wages reflects Japanese firms continuing to limit wage growth in the goods sector, thereby preventing positive spillovers to workers in the services sector. In other words, rather than a reverse Balassa-Samuelson effect, which can be blamed on sluggish productivity growth, the Balassa-Samuelson effect has been obstructed such that productivity improvements have not boosted wages (and, as a result, JPY REER has kept sliding).

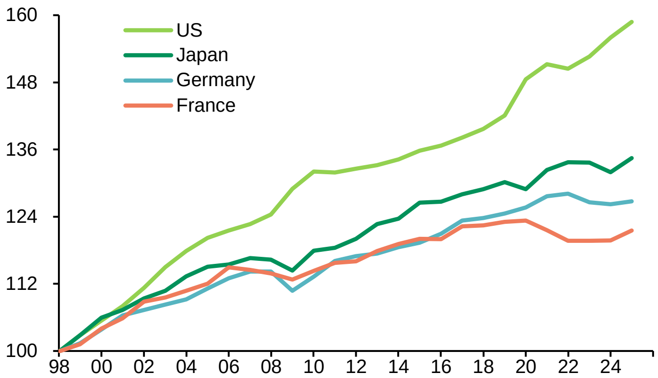
Japan's productivity growth has not translated into higher wages: The yen's real buying power relative to the USD has fallen by 55% since 1998, while the euro's real buying power in Germany and France has declined by 11% and 34%, respectively (Figure 5). These differences reflect not only nominal exchange rates but also differing inflation rates across the three countries.

Japan's wage level (per capita, full-time equivalent, and nominal) relative to the US has plunged 58% during the same period, compared with a 21% fall in Germany and a 22% decrease in France (Figure 6). Meanwhile, the relative price level has dropped 44% in Japan, 18% in Germany, and 23% in France (Figure 7).

The US indeed appears to be benefiting from the Balassa-Samuelson effect, given significant productivity improvements there. During the same time, Japan's real exchange rate has weakened by much more than Germany's or France's, despite Japan's superior productivity growth and sharp declines in relative wage and price levels, indicating that Japan's productivity growth has not translated into higher wages.

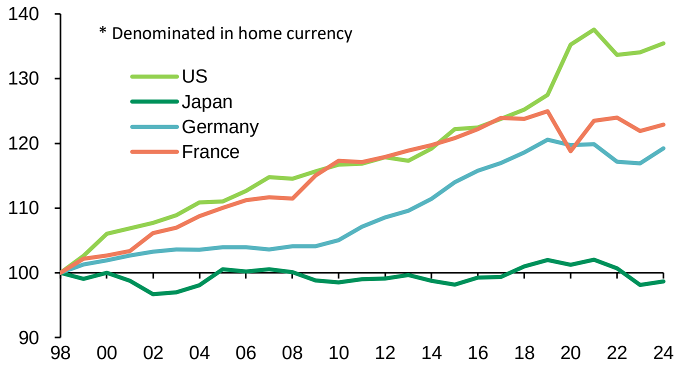
Declines in relative wage and price levels have eroded the yen's real buying power: Since 1998, Japan's real exchange rate has weakened by 50% against Germany and 46% against France. Over the same period, Japan's wages fell by 47% versus Germany and 46% versus France, while its price levels eased by 31% and 27%, respectively. All of this has happened despite Japan outpacing both Germany and France in productivity growth, suggesting Japan's productivity gains have not translated into stronger wage growth.

Fig. 3: Hourly productivity (1998 = 100)



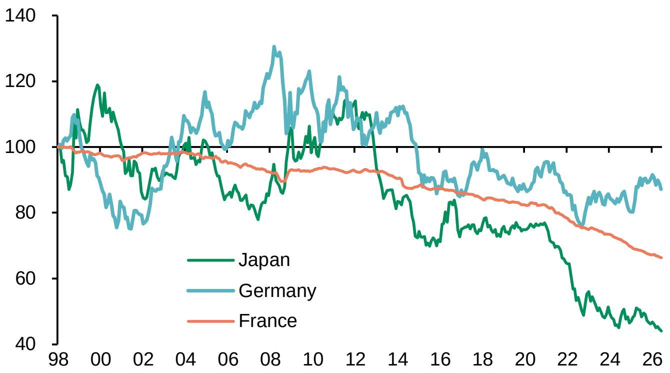
Sources: Macrobond, OECD, MIC, MHLW, Cabinet office Japan, BNP Paribas

Fig. 4: Real wages (1998 = 100, full-time equivalent)



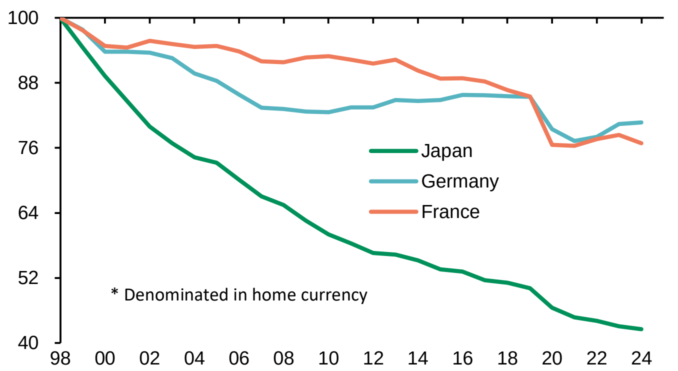
Sources: Macrobond, OECD, BNP Paribas

Fig. 5: Real effective exchange rate relative to USD (1998 = 100)



Sources: Macrobond, BNP Paribas

Fig. 6: Relative wage level versus US (Nominal, per worker, and full-time equivalent; 1998 = 100)



Sources: Macrobond, OECD, BNP Paribas

Ideally, we should distinguish between goods and services when examining the Balassa-Samuelson effect. Still, at least at the macroeconomic level, we believe it is difficult to make a strong case for Japan being a productivity laggard when compared with Germany and France.

The appropriately titled *Balassa-Samuelson in the long run: Qualitative success, quantitative limits* paper, published by Shinnosuke Kikuchi of the University of California (San Diego) this year, shows that while the data does offer qualitative support for the existence of the Balassa-Samuelson effect "on average across countries", "quantitatively … a standard multi-country trade model fed only by observed sectoral productivity cannot match country paths and delivers too small magnitudes."

BoJ is not solely responsible for the yen's weakness: The BoJ's gradual monetary policy normalisation has been a negative for the yen. Exchange rates, of course, tend to be significantly impacted by monetary policy and interest rate differentials in the short to medium term. But the yen has now been weakening for more than a quarter of a century, reflecting the structural downward pressure associated with declines in Japan's relative wage and price levels, as domestic firms have repeatedly refused to reward workers for their improving productivity.

The reasons for the yen having depreciated so significantly since global inflation took off in 2022 may also need to be reconsidered. Many have pointed to the BoJ's insistence on sticking to its unprecedented monetary easing framework even as the Fed and European central banks quickened their rate hikes, but here, too, blame needs to be placed on Japanese firms for not having matched the big wage increases offered by their US and European counterparts (with the net result that declines in Japan's relative wage and price levels have accelerated).

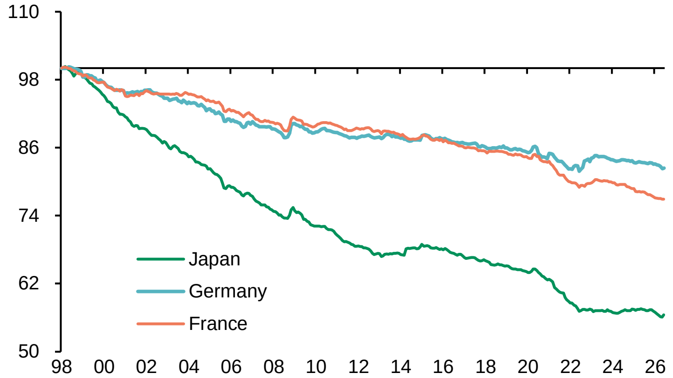
Yen weakness has barely been corrected even as real interest rate differentials have narrowed: It could conceivably be a different story going forward if Japan finds itself alone in experiencing high wage growth and inflation, but absent such a scenario, we do not expect more than a modest appreciation of the yen in real terms, even if interest rate differentials versus the US and Europe narrow due to continued BoJ monetary policy normalisation.

Japanese businesses and labour unions need to take this situation seriously. Japan's industrial competitiveness has indeed been bolstered over the past 25 years by the efforts made to prevent labour costs from exerting upward pressure on export prices, thereby enabling firms to maintain strong earnings since the 2010s even while preserving the long-term employment system. But the typical Japanese citizen has paid a heavy price for this in significantly reduced purchasing power.

Japan is now an extremely cheap place to visit even for tourists from emerging economies, while many citizens can now do little more than dream about travelling overseas. Booming inbound consumption is merely a reflection of foreign tourists taking full advantage of cheap Japanese labour and should certainly not be viewed as a 'success story' by the government, the businesses, and labour unions. Even many countries with lower productivity than Japan have seen nominal wages double over the past 25 years (reflecting annual growth of around +3%), so business leaders and labour unions would be ill-advised to rest on their laurels after having agreed to base-pay increases slightly north of 3% for three straight years since 2024, in our view.

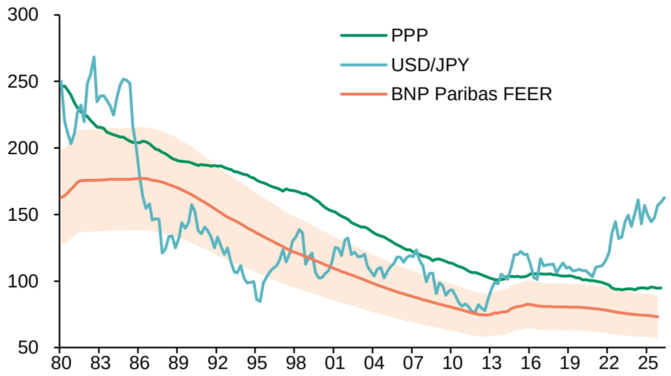
The yen's nominal exchange rate and PPP have diverged: Real exchange rates have risen over the past quarter of a century for nations with higher inflation rates while falling for those with milder inflation. This is completely at odds with purchasing power parity (PPP) theory.

Fig. 7: Relative price level versus US CPI (1998 = 100)



Sources: Macrobond, BNP Paribas

Fig. 8: PPP, USDJPY, and BNP Paribas FEER*



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Sources: Macrobond, OECD, BNP Paribas

The (relative) PPP hypothesis contends that – all else being equal – a country experiencing high inflation should see its nominal exchange rate weaken commensurately to hold the real exchange rate steady. However, the yen has continued to weaken since around the turn of the century even against a backdrop of persistently low inflation.

The nominal USDJPY exchange rate was around 100 in the early 2000s, but the OECD estimates that the theoretically fair level consistent with PPP at that time was a little south of 160 (Figure 8). We have seen a complete turnaround since then, with USDJPY now at around 160 (meaning that the yen is much weaker in nominal terms) and the OECD's PPP estimate at roughly 95 (suggesting that the yen should theoretically be much stronger).

Here too, obstruction of the Balassa-Samuelson effect seems to have been a major factor. On the PPP side, the yen has ‘strengthened’ due to neither wages nor prices rising even as productivity kept improving. At the same time, the nominal exchange rate has continued diverging from its theoretically fair level as a result of persistent declines in Japan's relative wage and price levels (and commensurate weakening of the yen in real terms).

One would ordinarily and quite reasonably expect wages to rise when productivity is improving, but that assumption has not held up in Japan's case, which is why neither the Balassa-Samuelson effect nor PPP ultimately offers an adequate explanation for the long-term trajectory of the yen.

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BNPP understand the importance of protecting confidential information and maintain a “need to know” approach when dealing with any confidential information. Information barriers are a key arrangement we have in place in this regard. Such arrangements, along with embedded policies and procedures, provide that information held in the course of carrying on one part of its business to be withheld from and not to be used in the course of carrying on another part of its business. It is a way of managing conflicts of interest whereby the business of the bank is separated by physical and non-physical information barriers. The Control Room manages this information flow between different areas of the bank where confidential information including inside information and proprietary information is safeguarded. There is also a conflict clearance process before getting involved in a deal or transaction.

In addition, there is a mitigation measure to manage conflicts of interest for each transaction with controls put in place to restrict the information flow, involvement of personnel and handling of client relations between each transaction in such a way that the different interests are appropriately protected. Gifts and Entertainment policy is to monitor physical gifts, benefits and invitation to events that is in line with the firm policy and Anti-Bribery regulations. BNPP maintains several policies with respect to conflicts of interest including our Personal Account Dealing and Outside Business Interests policies which sit alongside our general Conflicts of Interest Policy, along with several policies that the firm has in place to prevent and avoid conflicts of interest.

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